VGG16 Training Summary ✅

Night and day difference after the fix! Here are the results:

Model	Dataset	Test Accuracy	Early Stop	Epochs
VGG16-BN (scratch)	ASL Commands (3 classes)	**100.00%**	Epoch 28	~24 min
VGG16-BN (scratch)	ASL Digits (10 classes)	**96.19%**	Epoch 28	~5 min
VGG16-BN (scratch)	ASL Alphabets (26 classes)	**99.95%**	Epoch 27	~100 min
VGG16 (transfer learning)	ASL Alphabets (26 classes)	**98.92%**	Ran all 30	~38 min

Key Highlights

- **Alphabets from scratch (99.95%)** is the best result across *all* your models so far — only **2 images** misclassified out of 3,900! Nearly every letter scored perfect 1.0000 F1.
- **Digits (96.19%)** also improved significantly — previously it was stuck at 10% (random chance).
- **From-scratch outperformed transfer learning again** (99.95% vs 98.92%) on Alphabets, confirming that for this ASL domain, learning features from scratch works better than adapting ImageNet features.

Terminal output:

PS C:\Users\samyb\Downloads\ai-models> py c:\Users\samyb\Downloads\ai-models\notebooks\VGG16_ASL.py

```
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB

============================================================
  VGG16 — ASL Commands
============================================================
  Train: 7200 images | Val: 900 | Test: 900
  Classes: ['del', 'nothing', 'space']
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
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Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
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VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

```
Epoch  1/30 — loss: 4.4343  acc: 0.6169  val_loss: 0.4500  val_acc: 0.8633  [83.2s]
Epoch  2/30 — loss: 1.6988  acc: 0.8043  val_loss: 0.2440  val_acc: 0.9478  [52.0s]
Epoch  3/30 — loss: 0.8364  acc: 0.8889  val_loss: 0.0423  val_acc: 0.9867  [52.3s]
Epoch  4/30 — loss: 0.4419  acc: 0.9369  val_loss: 0.0084  val_acc: 0.9967  [52.1s]
Epoch  5/30 — loss: 0.2757  acc: 0.9557  val_loss: 0.0002  val_acc: 1.0000  [52.1s]
Epoch  6/30 — loss: 0.1620  acc: 0.9724  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
Epoch  7/30 — loss: 0.0809  acc: 0.9836  val_loss: 0.0003  val_acc: 1.0000  [52.1s]
Epoch  8/30 — loss: 0.1219  acc: 0.9815  val_loss: 0.0001  val_acc: 1.0000  [52.1s]
Epoch  9/30 — loss: 0.1057  acc: 0.9826  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
Epoch 10/30 — loss: 0.0675  acc: 0.9857  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
Epoch 11/30 — loss: 0.0578  acc: 0.9910  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
Epoch 12/30 — loss: 0.0509  acc: 0.9921  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
Epoch 13/30 — loss: 0.0184  acc: 0.9960  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
Epoch 14/30 — loss: 0.0233  acc: 0.9958  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
Epoch 15/30 — loss: 0.0416  acc: 0.9917  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
Epoch 16/30 — loss: 0.0398  acc: 0.9931  val_loss: 0.0154  val_acc: 0.9978  [52.0s]
Epoch 17/30 — loss: 0.0357  acc: 0.9938  val_loss: 0.0018  val_acc: 0.9989  [52.1s]
Epoch 18/30 — loss: 0.0294  acc: 0.9957  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
Epoch 19/30 — loss: 0.0251  acc: 0.9958  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
```

```
  Epoch 20/30 — loss: 0.0218  acc: 0.9953  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
  Epoch 21/30 — loss: 0.0292  acc: 0.9942  val_loss: 0.0000  val_acc: 1.0000  [52.5s]
  Epoch 22/30 — loss: 0.0294  acc: 0.9950  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
  Epoch 23/30 — loss: 0.0218  acc: 0.9961  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
  Epoch 24/30 — loss: 0.0134  acc: 0.9972  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
  Epoch 25/30 — loss: 0.0134  acc: 0.9974  val_loss: 0.0000  val_acc: 1.0000  [52.0s]
  Epoch 26/30 — loss: 0.0190  acc: 0.9965  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
  Epoch 27/30 — loss: 0.0326  acc: 0.9953  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
  Epoch 28/30 — loss: 0.0070  acc: 0.9986  val_loss: 0.0000  val_acc: 1.0000  [52.1s]
  Early stopping at epoch 28 (patience=5)
  Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\commands_history.png
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB

============================================================
  Classification Report — VGG16 — ASL Commands
```

```
============================================================
              precision    recall  f1-score   support

         del     1.0000    1.0000    1.0000       300
     nothing     1.0000    1.0000    1.0000       300
       space     1.0000    1.0000    1.0000       300

    accuracy                         1.0000       900
   macro avg     1.0000    1.0000    1.0000       900
weighted avg     1.0000    1.0000    1.0000       900

  Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\commands_confusion.png

  Test Accuracy: 1.0000
  ✓ Model saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\commands_model.pth

============================================================
  VGG16 — ASL Digits
============================================================
  Train: 1644 images | Val: 208 | Test: 210
  Classes: ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9']
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
```

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GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
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Device: cuda
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VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Epoch 1/30 — loss: 9.3700 acc: 0.1046 val_loss: 4.5129 val_acc: 0.1202 [40.6s]
Epoch 2/30 — loss: 8.4910 acc: 0.1399 val_loss: 2.2190 val_acc: 0.2500 [12.1s]
Epoch 3/30 — loss: 7.2057 acc: 0.1727 val_loss: 1.7958 val_acc: 0.4663 [12.1s]
Epoch 4/30 — loss: 6.1072 acc: 0.2403 val_loss: 1.4589 val_acc: 0.5433 [12.1s]
Epoch 5/30 — loss: 5.2485 acc: 0.3005 val_loss: 1.2976 val_acc: 0.5913 [12.1s]
Epoch 6/30 — loss: 4.6604 acc: 0.3449 val_loss: 1.0625 val_acc: 0.7500 [12.1s]
```

```
Epoch  7/30 — loss: 3.9303  acc: 0.4118  val_loss: 1.0232  val_acc: 0.6827  [12.1s]
Epoch  8/30 — loss: 3.3336  acc: 0.4848  val_loss: 0.6944  val_acc: 0.7740  [12.1s]
Epoch  9/30 — loss: 3.0440  acc: 0.5122  val_loss: 0.4672  val_acc: 0.8558  [12.1s]
Epoch 10/30 — loss: 2.5542  acc: 0.5639  val_loss: 0.4408  val_acc: 0.8750  [12.1s]
Epoch 11/30 — loss: 2.2600  acc: 0.5985  val_loss: 0.5098  val_acc: 0.8413  [12.1s]
Epoch 12/30 — loss: 2.2667  acc: 0.6046  val_loss: 0.3570  val_acc: 0.8942  [12.2s]
Epoch 13/30 — loss: 1.9788  acc: 0.6338  val_loss: 0.3320  val_acc: 0.9038  [12.1s]
Epoch 14/30 — loss: 2.0109  acc: 0.6466  val_loss: 0.2109  val_acc: 0.9519  [12.1s]
Epoch 15/30 — loss: 1.4884  acc: 0.7007  val_loss: 0.4354  val_acc: 0.8654  [12.1s]
Epoch 16/30 — loss: 1.4168  acc: 0.7086  val_loss: 0.2367  val_acc: 0.9375  [12.1s]
Epoch 17/30 — loss: 1.3741  acc: 0.7147  val_loss: 0.2132  val_acc: 0.9423  [12.1s]
Epoch 18/30 — loss: 1.3291  acc: 0.7293  val_loss: 0.2084  val_acc: 0.9423  [12.1s]
Epoch 19/30 — loss: 1.2483  acc: 0.7372  val_loss: 0.1702  val_acc: 0.9471  [12.1s]
Epoch 20/30 — loss: 1.1064  acc: 0.7488  val_loss: 0.2701  val_acc: 0.9327  [12.1s]
Epoch 21/30 — loss: 0.9440  acc: 0.7841  val_loss: 0.1906  val_acc: 0.9423  [12.2s]
Epoch 22/30 — loss: 0.9184  acc: 0.7749  val_loss: 0.2936  val_acc: 0.9423  [12.1s]
Epoch 23/30 — loss: 0.9326  acc: 0.7871  val_loss: 0.1257  val_acc: 0.9663  [12.1s]
Epoch 24/30 — loss: 0.8060  acc: 0.8023  val_loss: 0.2258  val_acc: 0.9375  [12.1s]
Epoch 25/30 — loss: 0.8642  acc: 0.8017  val_loss: 0.1818  val_acc: 0.9423  [12.1s]
Epoch 26/30 — loss: 0.8097  acc: 0.8120  val_loss: 0.2409  val_acc: 0.9423  [12.1s]
Epoch 27/30 — loss: 0.7260  acc: 0.8163  val_loss: 0.1378  val_acc: 0.9712  [12.1s]
Epoch 28/30 — loss: 0.6719  acc: 0.8345  val_loss: 0.1979  val_acc: 0.9567  [12.1s]
Early stopping at epoch 28 (patience=5)
Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\digits_history.png
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
```

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

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Classification Report — VGG16 — ASL Digits

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	precision	recall	f1-score	support
0	1.0000	1.0000	1.0000	21
1	1.0000	1.0000	1.0000	21
2	1.0000	1.0000	1.0000	21
3	0.9524	0.9524	0.9524	21
4	0.8750	1.0000	0.9333	21
5	1.0000	1.0000	1.0000	21
6	1.0000	0.8571	0.9231	21
7	0.9474	0.8571	0.9000	21
8	0.9130	1.0000	0.9545	21
9	0.9524	0.9524	0.9524	21

```
   accuracy                     0.9619      210
  macro avg    0.9640    0.9619    0.9616      210
weighted avg    0.9640    0.9619    0.9616      210

 Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\digits_confusion.png

 Test Accuracy: 0.9619
 ✓ Model saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\digits_model.pth

============================================================
 VGG16 — ASL Alphabets
============================================================
 Train: 31200 images | Val: 3900 | Test: 3900
 Classes: ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X',
'Y', 'Z']
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
```

```
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Epoch 1/30 — loss: 5.4224 acc: 0.0722 val_loss: 4.0813 val_acc: 0.2354 [253.5s]
Epoch 2/30 — loss: 3.8326 acc: 0.1547 val_loss: 1.9463 val_acc: 0.4282 [224.5s]
Epoch 3/30 — loss: 2.8976 acc: 0.3087 val_loss: 2.3933 val_acc: 0.5482 [224.4s]
Epoch 4/30 — loss: 1.9717 acc: 0.4722 val_loss: 0.6491 val_acc: 0.7946 [224.5s]
Epoch 5/30 — loss: 1.1310 acc: 0.6431 val_loss: 0.4217 val_acc: 0.8569 [224.3s]
Epoch 6/30 — loss: 0.7198 acc: 0.7596 val_loss: 0.1703 val_acc: 0.9472 [224.4s]
Epoch 7/30 — loss: 0.5049 acc: 0.8309 val_loss: 0.0859 val_acc: 0.9726 [224.5s]
Epoch 8/30 — loss: 0.3506 acc: 0.8834 val_loss: 0.0941 val_acc: 0.9695 [224.8s]
Epoch 9/30 — loss: 0.2659 acc: 0.9116 val_loss: 0.0383 val_acc: 0.9890 [224.5s]
Epoch 10/30 — loss: 0.2152 acc: 0.9300 val_loss: 0.0538 val_acc: 0.9800 [224.4s]
Epoch 11/30 — loss: 0.1680 acc: 0.9449 val_loss: 0.0300 val_acc: 0.9895 [224.5s]
Epoch 12/30 — loss: 0.1467 acc: 0.9526 val_loss: 0.0419 val_acc: 0.9872 [226.6s]
```

```
Epoch 13/30 — loss: 0.1214  acc: 0.9608  val_loss: 0.0150  val_acc: 0.9964  [226.6s]
Epoch 14/30 — loss: 0.1049  acc: 0.9675  val_loss: 0.0214  val_acc: 0.9928  [226.6s]
Epoch 15/30 — loss: 0.0997  acc: 0.9682  val_loss: 0.0118  val_acc: 0.9964  [225.1s]
Epoch 16/30 — loss: 0.0839  acc: 0.9742  val_loss: 0.0139  val_acc: 0.9946  [224.5s]
Epoch 17/30 — loss: 0.0764  acc: 0.9763  val_loss: 0.0131  val_acc: 0.9977  [224.5s]
Epoch 18/30 — loss: 0.0708  acc: 0.9779  val_loss: 0.0091  val_acc: 0.9969  [224.5s]
Epoch 19/30 — loss: 0.0631  acc: 0.9798  val_loss: 0.0252  val_acc: 0.9921  [224.3s]
Epoch 20/30 — loss: 0.0583  acc: 0.9812  val_loss: 0.0029  val_acc: 0.9990  [223.8s]
Epoch 21/30 — loss: 0.0483  acc: 0.9846  val_loss: 0.0033  val_acc: 0.9985  [224.5s]
Epoch 22/30 — loss: 0.0575  acc: 0.9821  val_loss: 0.0010  val_acc: 1.0000  [224.5s]
Epoch 23/30 — loss: 0.0466  acc: 0.9854  val_loss: 0.0027  val_acc: 0.9987  [224.6s]
Epoch 24/30 — loss: 0.0480  acc: 0.9854  val_loss: 0.0213  val_acc: 0.9944  [224.6s]
Epoch 25/30 — loss: 0.0406  acc: 0.9873  val_loss: 0.0024  val_acc: 0.9995  [224.9s]
Epoch 26/30 — loss: 0.0356  acc: 0.9899  val_loss: 0.0016  val_acc: 0.9995  [224.5s]
Epoch 27/30 — loss: 0.0297  acc: 0.9908  val_loss: 0.0033  val_acc: 0.9990  [224.4s]
Early stopping at epoch 27 (patience=5)
Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\alphabets_history.png
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
```

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

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Classification Report — VGG16 — ASL Alphabets

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	precision	recall	f1-score	support
A	1.0000	1.0000	1.0000	150
B	1.0000	1.0000	1.0000	150
C	1.0000	0.9933	0.9967	150
D	1.0000	1.0000	1.0000	150
E	1.0000	1.0000	1.0000	150
F	1.0000	1.0000	1.0000	150
G	1.0000	1.0000	1.0000	150
H	1.0000	1.0000	1.0000	150
I	1.0000	1.0000	1.0000	150
J	1.0000	1.0000	1.0000	150
K	1.0000	1.0000	1.0000	150
L	1.0000	1.0000	1.0000	150
M	1.0000	1.0000	1.0000	150
N	1.0000	1.0000	1.0000	150
O	0.9934	1.0000	0.9967	150
P	1.0000	1.0000	1.0000	150
Q	1.0000	1.0000	1.0000	150

```
           R    0.9934    1.0000    0.9967       150
           S    1.0000    1.0000    1.0000       150
           T    1.0000    1.0000    1.0000       150
           U    1.0000    0.9933    0.9967       150
           V    1.0000    1.0000    1.0000       150
           W    1.0000    1.0000    1.0000       150
           X    1.0000    1.0000    1.0000       150
           Y    1.0000    1.0000    1.0000       150
           Z    1.0000    1.0000    1.0000       150

    accuracy                         0.9995      3900
   macro avg     0.9995    0.9995    0.9995      3900
weighted avg     0.9995    0.9995    0.9995      3900

  Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\alphabets_confusion.png

  Test Accuracy: 0.9995
  ✓ Model saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\alphabets_model.pth

============================================================
  VGG16 Transfer Learning — ASL Alphabets
============================================================
  Train: 31200 images | Val: 3900 | Test: 3900
  Classes: ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z']
```

Downloading: "https://download.pytorch.org/models/vgg16-397923af.pth" to C:\Users\samyb/.cache\torch\hub\checkpoints\vgg16-397923af.pth

100%|[redacted]| 528M/528M [01:04<00:00, 8.52MB/s]

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

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Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

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Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

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Device: cuda

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VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

Device: cuda

GPU: NVIDIA GeForce RTX 4060 Laptop GPU

VRAM: 8.0 GB

```
Epoch  1/30 — loss: 1.8871  acc: 0.5635  val_loss: 0.9328  val_acc: 0.8959  [105.9s]
Epoch  2/30 — loss: 0.7699  acc: 0.8526  val_loss: 0.4090  val_acc: 0.9472  [76.9s]
Epoch  3/30 — loss: 0.4697  acc: 0.8990  val_loss: 0.2468  val_acc: 0.9638  [76.6s]
Epoch  4/30 — loss: 0.3462  acc: 0.9175  val_loss: 0.1862  val_acc: 0.9687  [76.5s]
Epoch  5/30 — loss: 0.2771  acc: 0.9310  val_loss: 0.1483  val_acc: 0.9756  [76.5s]
Epoch  6/30 — loss: 0.2452  acc: 0.9337  val_loss: 0.1372  val_acc: 0.9685  [76.6s]
Epoch  7/30 — loss: 0.2148  acc: 0.9422  val_loss: 0.1209  val_acc: 0.9703  [76.6s]
Epoch  8/30 — loss: 0.1956  acc: 0.9460  val_loss: 0.0892  val_acc: 0.9851  [76.6s]
Epoch  9/30 — loss: 0.1822  acc: 0.9473  val_loss: 0.0814  val_acc: 0.9810  [76.6s]
Epoch 10/30 — loss: 0.1695  acc: 0.9508  val_loss: 0.0801  val_acc: 0.9826  [76.5s]
Epoch 11/30 — loss: 0.1621  acc: 0.9526  val_loss: 0.0732  val_acc: 0.9831  [76.6s]
Epoch 12/30 — loss: 0.1500  acc: 0.9551  val_loss: 0.0820  val_acc: 0.9744  [76.6s]
Epoch 13/30 — loss: 0.1373  acc: 0.9587  val_loss: 0.0592  val_acc: 0.9864  [76.5s]
Epoch 14/30 — loss: 0.1354  acc: 0.9595  val_loss: 0.0605  val_acc: 0.9864  [76.5s]
Epoch 15/30 — loss: 0.1313  acc: 0.9604  val_loss: 0.0479  val_acc: 0.9887  [76.9s]
Epoch 16/30 — loss: 0.1290  acc: 0.9597  val_loss: 0.0566  val_acc: 0.9838  [76.6s]
Epoch 17/30 — loss: 0.1219  acc: 0.9623  val_loss: 0.0495  val_acc: 0.9856  [76.6s]
Epoch 18/30 — loss: 0.1129  acc: 0.9647  val_loss: 0.0454  val_acc: 0.9882  [76.5s]
Epoch 19/30 — loss: 0.1145  acc: 0.9635  val_loss: 0.0484  val_acc: 0.9869  [76.5s]
Epoch 20/30 — loss: 0.1106  acc: 0.9659  val_loss: 0.0438  val_acc: 0.9903  [76.6s]
Epoch 21/30 — loss: 0.1072  acc: 0.9664  val_loss: 0.0420  val_acc: 0.9882  [76.5s]
Epoch 22/30 — loss: 0.1094  acc: 0.9658  val_loss: 0.0431  val_acc: 0.9877  [76.6s]
Epoch 23/30 — loss: 0.1020  acc: 0.9667  val_loss: 0.0387  val_acc: 0.9910  [76.5s]
Epoch 24/30 — loss: 0.1011  acc: 0.9678  val_loss: 0.0427  val_acc: 0.9887  [76.6s]
Epoch 25/30 — loss: 0.0987  acc: 0.9681  val_loss: 0.0377  val_acc: 0.9892  [76.6s]
```

```
  Epoch 26/30 — loss: 0.0996  acc: 0.9682  val_loss: 0.0309  val_acc: 0.9931  [76.6s]
  Epoch 27/30 — loss: 0.0941  acc: 0.9707  val_loss: 0.0417  val_acc: 0.9885  [76.9s]
  Epoch 28/30 — loss: 0.0904  acc: 0.9719  val_loss: 0.0371  val_acc: 0.9905  [76.6s]
  Epoch 29/30 — loss: 0.0912  acc: 0.9710  val_loss: 0.0344  val_acc: 0.9903  [76.6s]
  Epoch 30/30 — loss: 0.0904  acc: 0.9704  val_loss: 0.0293  val_acc: 0.9931  [76.5s]
  Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\alphabets_tl_history.png
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB
Device: cuda
GPU: NVIDIA GeForce RTX 4060 Laptop GPU
VRAM: 8.0 GB

============================================================
  Classification Report — VGG16 Transfer Learning — ASL Alphabets
============================================================
              precision    recall  f1-score   support

           A     0.9933    0.9867    0.9900       150
           B     1.0000    1.0000    1.0000       150
```

C	1.0000	1.0000	1.0000	150
D	1.0000	1.0000	1.0000	150
E	0.9675	0.9933	0.9803	150
F	1.0000	1.0000	1.0000	150
G	0.9868	0.9933	0.9900	150
H	1.0000	0.9867	0.9933	150
I	1.0000	0.9800	0.9899	150
J	0.9868	1.0000	0.9934	150
K	0.9868	1.0000	0.9934	150
L	1.0000	1.0000	1.0000	150
M	0.9615	1.0000	0.9804	150
N	1.0000	0.9600	0.9796	150
O	1.0000	1.0000	1.0000	150
P	1.0000	1.0000	1.0000	150
Q	1.0000	1.0000	1.0000	150
R	0.9669	0.9733	0.9701	150
S	0.9367	0.9867	0.9610	150
T	0.9799	0.9733	0.9766	150
U	0.9669	0.9733	0.9701	150
V	1.0000	0.9800	0.9899	150
W	1.0000	1.0000	1.0000	150
X	1.0000	0.9533	0.9761	150
Y	0.9933	0.9867	0.9900	150
Z	1.0000	0.9933	0.9967	150
accuracy		0.9892	3900	

```
   macro avg     0.9895    0.9892    0.9893      3900
weighted avg     0.9895    0.9892    0.9893      3900

Saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\alphabets_tl_confusion.png

Test Accuracy: 0.9892
✓ Model saved: c:\Users\samyb\Downloads\ai-models\results\vgg16\alphabets_tl_model.pth

✓ All done! Results saved to: c:\Users\samyb\Downloads\ai-models\results\vgg16
```